

Assessment Tool-1

CO1 - Analytical Problem

Reg: 192525 468

Name: P. Devendra Reddy

Solving

Coin Toss - Maximum Likelihood Estimation.

Given:

• Total Tosses = 10

• Outcomes = H H T H H H T H H T

• No. of Heads = 7

Formula:
$$\hat{p} = \frac{\text{No. of Heads}}{\text{Total Observations}}$$

$$\hat{p} = \frac{7}{10} = 0.7$$

The MLE Estimate of the probability of getting a Head is 0.7 (70%).

Q. Student pass probability using MLE

Given:

• Total Students = 50

• Students passed = 42

$$\hat{p} = \frac{\text{No. of Successes}}{\text{Total Trials}}$$

$$\hat{p} = \frac{42}{50}$$

$$= 0.84$$

The MLE estimate of the probability that a student passes is 0.84 (84%)

(3)

Defective Bulbs using MLE

Given

Total Bulbs = 100

Defective Bulbs = 8

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

$$\hat{p} = \frac{8}{100} = 0.08$$

The estimate probability that a bulb is defective 0.08 (8%)

(4)

Biased Die using MLE

Given:

Total Rolls = 60

Number of Sixes = 18

$$\hat{p} = \frac{\text{No. of Sides}}{\text{Total Rolls}}$$

$$\hat{p} = \frac{18}{60} = 0.3$$

The MLE estimate of rolling 6 is 0.3 (30%)

③ Gradient Descent problem.

$$\text{Input } x = 4$$

$$\text{weight } w = 2$$

$$\text{Actual output} = 10$$

$$\text{Learning Rate } \eta = 0.01$$

$$\text{Gradient} = 4$$

Formula:

$$y = w \times x$$

$$y = 2 \times 4 = 8$$

$$\text{Predicted output} = 8$$

Step 2: Error

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

$$= 10 - 8 = 2$$

$$\text{Error} = 2$$

Step 3: update weight

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04 = 1.96$$

$$\text{Updated weight} = 1.96 = \frac{81}{0.04} = 9$$

Conclusion: The predicted output is 8, The error is 2, and after one gradient descent update, The new weight becomes 1.96.